1b. The divide-by-n estimator is biased downward across the 1,000 samples of n = 5 it averaged 3.26, which is below the true variance of 4. The divide by (n − 1) estimator averaged 4.08, close to 4. The divide by n estimator is biased.

1d. The estimated bias of s is −0.269 (n = 3), −0.096 (n = 5), −0.070 (n = 10), −0.010 (n = 30), and +0.001 (n = 100). So s is biased downward for $\sigma = 2$ in small samples, even though s^2 is unbiased for σ^2, and the bias shrinks toward zero as n grows. At n = 100 it is practically zero, and the tiny positive value is irrelevant.

1e. At n = 5 the divide-by-n estimates have a standard deviation of 2.27 versus 2.84 for the divide by (n − 1) estimates, so the divide by n estimator is less variable. That doesn't make it better because it is centered on the wrong value (3.26 instead of 4).

2b. The model without z estimated the effect of x as 3.81, far above the true value of 2. The model with z estimated 2.18 for x and 2.86 for z, much closer to the true values of 2 and 3. In the plot, the purple line (without z) is very noticeably steeper than the red line (true slope), while the green line (with z) is closer to it. Omitting z biases x's coefficient upward because z affects y and is positively correlated with x, so x's coefficient picks up part of z's effect.

2c. The model without z still gives 3.53 for x, almost unchanged from 3.81 and still far from 2. The model with z gives 2.03 for x and 2.97 for z, which are essentially the true values. More data did not resolve the bias, it made estimates more stable, but the short model is still stable around the wrong number.

2d. The correlation between x and z was −0.03, essentially zero. Now both models give estimates near 2: 1.91 without z and 1.99 with z. Omitting variable z no longer causes meaningful bias, z still affects y. But, because it is unrelated to x, it isn't a confounder. Thus, leaving it out doesn't distort x's coefficient.